BRICS Elastic Monetary Architecture with Clearing-Based Activity Anchors and a Governance Reserve

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Abstract. We integrate (i) empirical evidence that aggregation smooths real-output volatility within the BRICS bloc and (ii) a rule-based monetary design for a basket unit featuring non-linear elastic reconvergence toward a macro fundamental. Governance is parameterized by a concentration dial λ (GDP^{λ} weights), with concentration measured by HHI and the dominant member share. We extend the baseline with stepwise reporting, stress tests, supply regimes (mirror burn β and an amortized 2% annual structural emission), and an implementable institutional layer where a clearing authority holds genesis supply S0 and executes mint/burn in a governance reserve (no automatic redistribution). We also show how clearing volumes define a monthly activity proxy (mini-GDP) for supply updates with smoothing and caps.

Keywords: BRICS; clearing unions; basket currencies; monetary design; stability; governance.

JEL: E42; E58; F33; F42.

1. Introduction

The practical problem addressed by this manuscript is institutional: how can a heterogeneous economic bloc build a settlement layer that is more stable than its members taken individually, while avoiding both full monetary union and excessive structural dominance by the largest economy. The approach is engineering-driven: we specify an explicit macro fundamental, a market-price component with noise, and a deterministic reconvergence rule with a closed-form local stability restriction. We then connect governance and resilience through a single dial (λ) and investigate stress behavior under discrete reporting and crisis-style volatility spikes. Finally, we translate the mechanism into an implementable clearing architecture with an authority-managed governance reserve.

2. Empirical evidence: stability of aggregated output

We summarize empirical evidence (2001-2024 growth rates derived from World Bank WDI real GDP in constant 2015 USD) showing that aggregated BRICS output growth is smoother than individual members. The core observation is volatility smoothing under aggregation and the role of weighting rules. We also provide an approximate 'BRICS without China' robustness diagnostic and a jackknife-style exclusion exercise based on second moments.

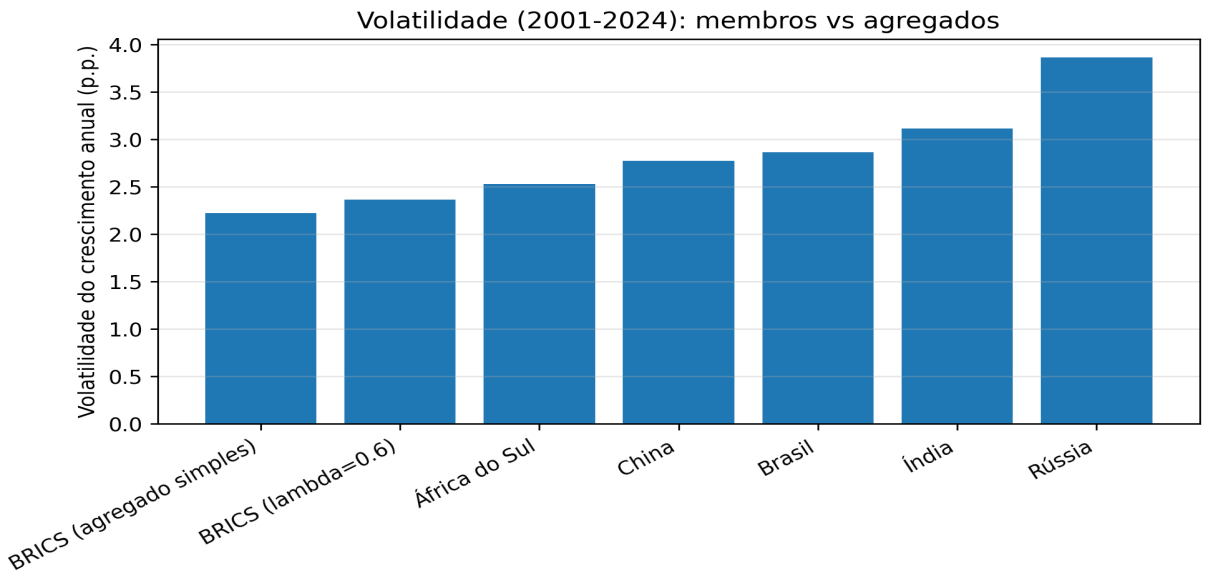

Figure 1. Empirical volatility of annual real GDP growth: members vs. aggregates.

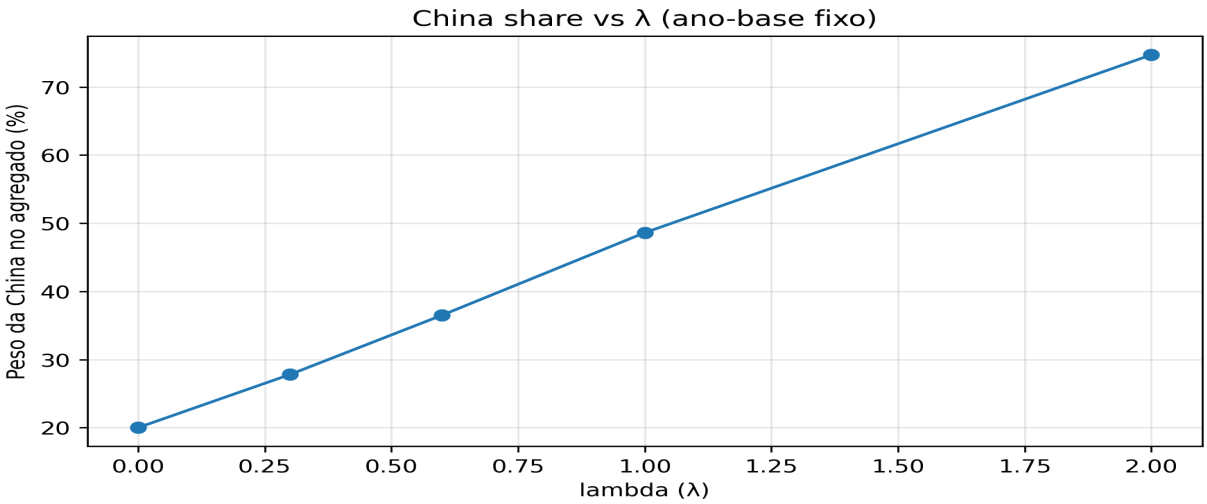

Figure 2. China share as a function of λ (empirical).

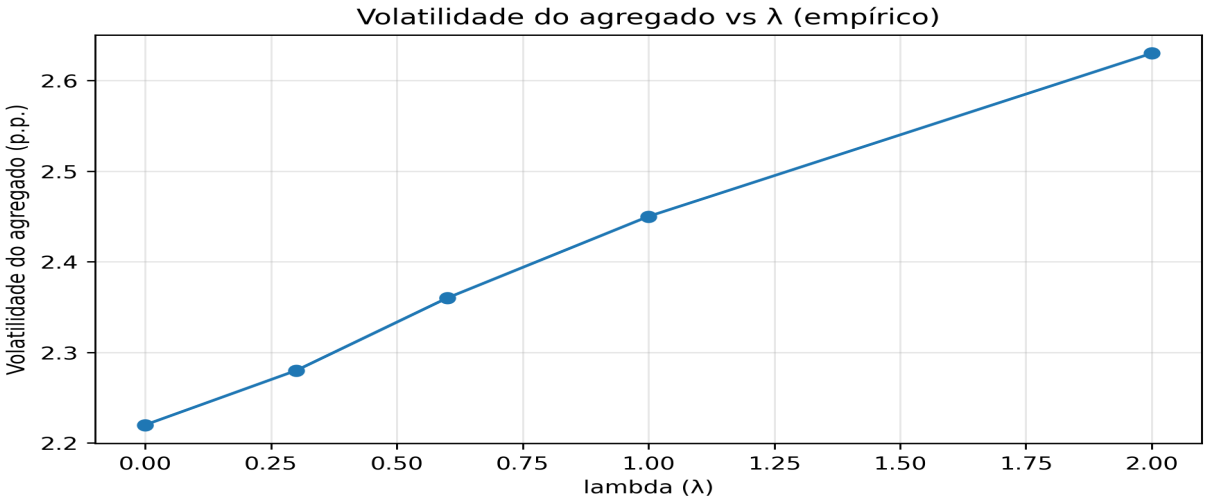

Figure 3. Aggregate volatility as a function of λ (empirical).

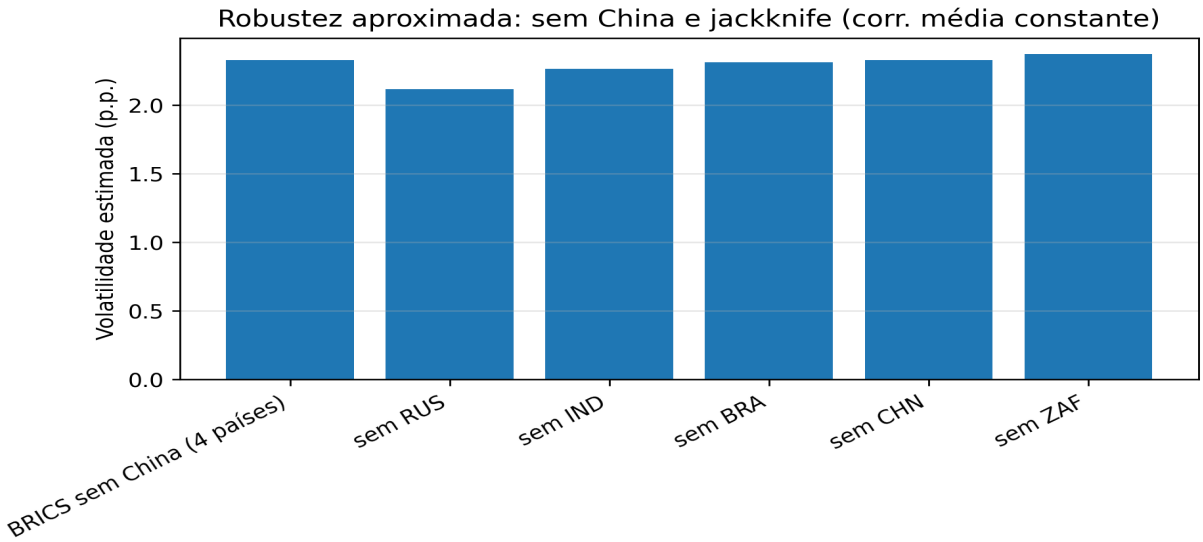

Figure 4. Robustness approximation: aggregate without China and jackknife (constant average correlation hypothesis).

3. Model: elastic reconvergence toward a macro fundamental

We define a macro fundamental as the ratio between a reported effective capital index and supply. Reported capital is updated in discrete windows to reflect real-world reporting frequency. Market price is subject to multiplicative speculation, and a non-linear reconvergence term corrects deviations.

$$F_t = \frac{\widehat{C}_t}{S_t}$$

$$P_t = (1-\alpha)P_{\text{spec},\,t} + \alpha\, P_{\text{spec},\,t}\left(\frac{F_t}{P_{\text{spec},\,t}}\right)^k$$

Local stability restriction (engineering constraint):

$$0 < \alpha k < 2$$

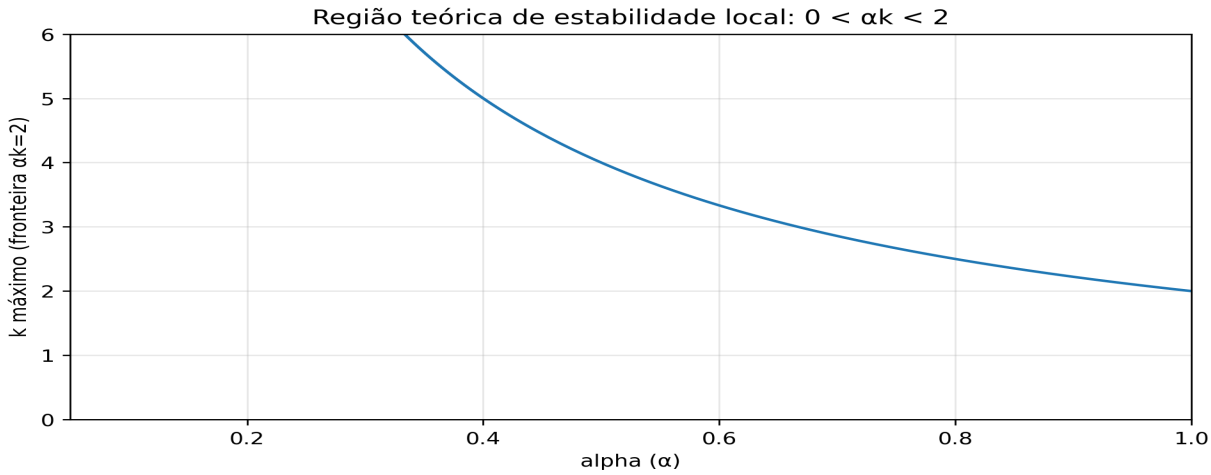

Figure 5. Local stability region: 0 < αk < 2.

3.1 Stepwise reporting ('staircase' fundamental)

To model institutional reporting frequency, we distinguish the latent daily evolution of capital from the reported anchor used by the monetary rule. The reported fundamental is updated on discrete windows (e.g., quarterly) and remains constant between releases. This produces characteristic plateaus with elastic re-anchoring.

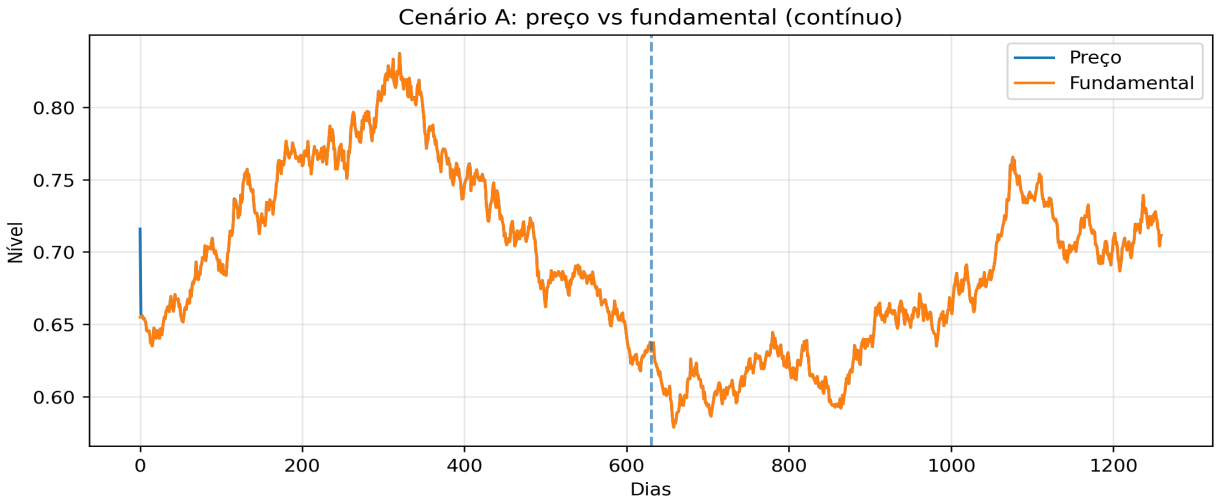

Figure 6. Scenario A: price vs. continuous fundamental (example path).

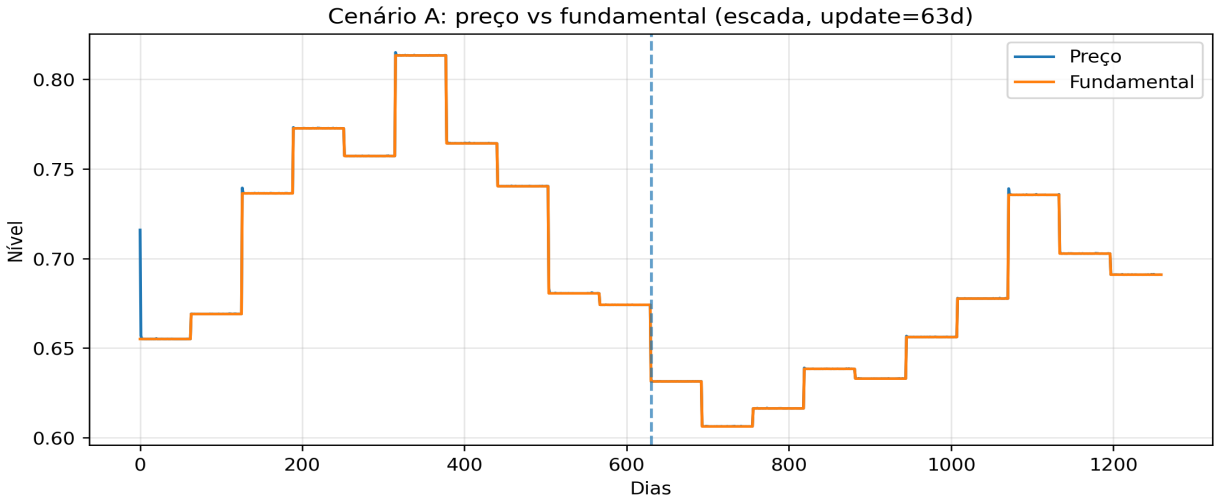

Figure 7. Scenario A: price vs. stepwise reported fundamental (quarterly update).

4. Governance and concentration: the λ parameter

Governance weights are parameterized by a power rule on economic size. The parameter λ controls structural concentration. Lower λ distributes weights more evenly; higher λ concentrates influence in the largest members. We quantify concentration using HHI and report the dominant member share.

$$w_i = \frac{\mathrm{GDP}_i^{\lambda}}{\sum_j \mathrm{GDP}_j^{\lambda}}$$

Table 1. Governance weights under λ = 0.5 (GDP proxy). HHI = 0.165.

Member	Weight (%)
CH	31.94
IN	14.48
RU	11.17
BR	10.91
SAU	7.90
IR	5.83
UAE	5.32
ZA	4.76
EG	4.76
ET	2.92

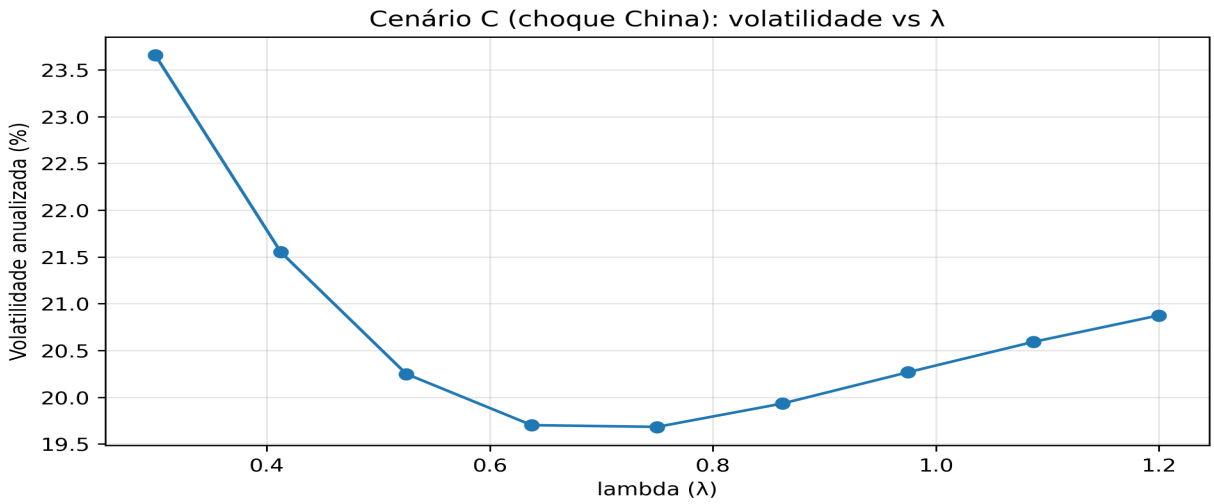

Figure 8. Scenario C (shock to dominant member): volatility vs. λ (Monte Carlo).

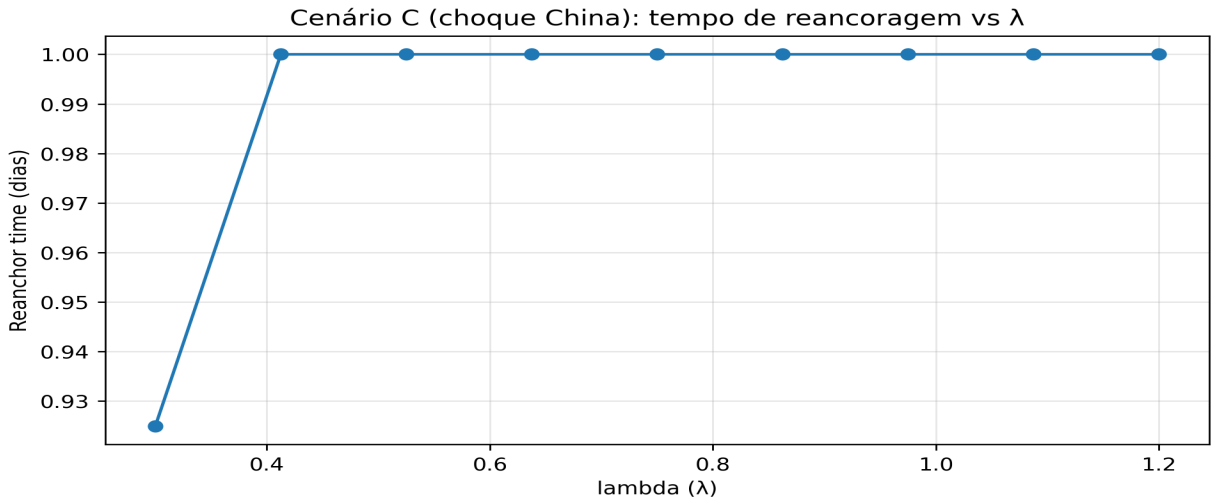

Figure 9. Scenario C: re-anchoring time (<= 5% band) vs. λ (Monte Carlo).

5. Stress tests

We evaluate two stress scenarios: (A) a macro shock applied to a subset of members and (B) a crisis-style scenario with a larger shock and a temporary volatility spike. Benchmarks include a macro GBM process. The figures report typical paths and summary risk metrics.

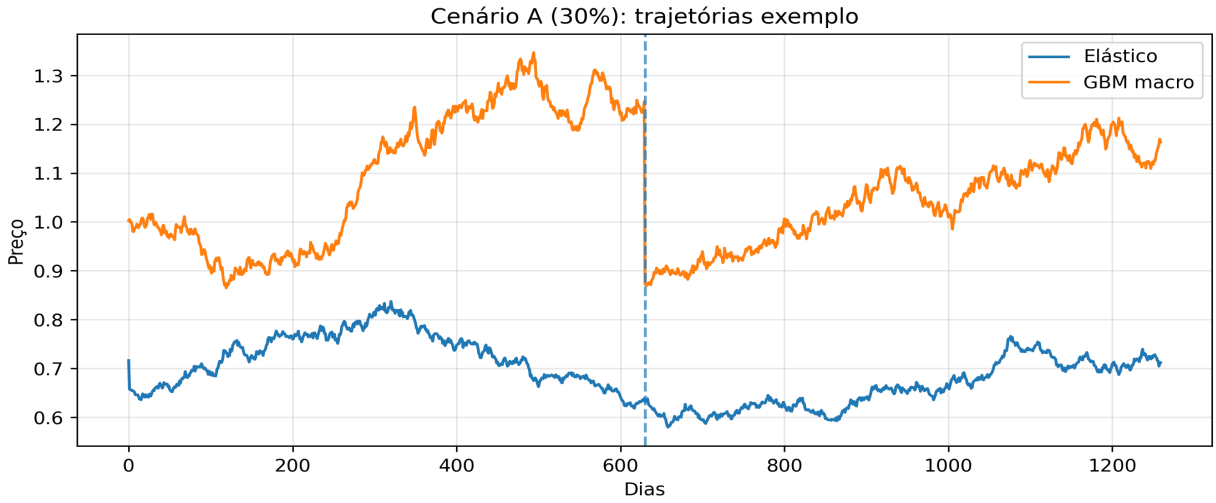

Figure 10. Scenario A: example trajectories (elastic vs. benchmark).

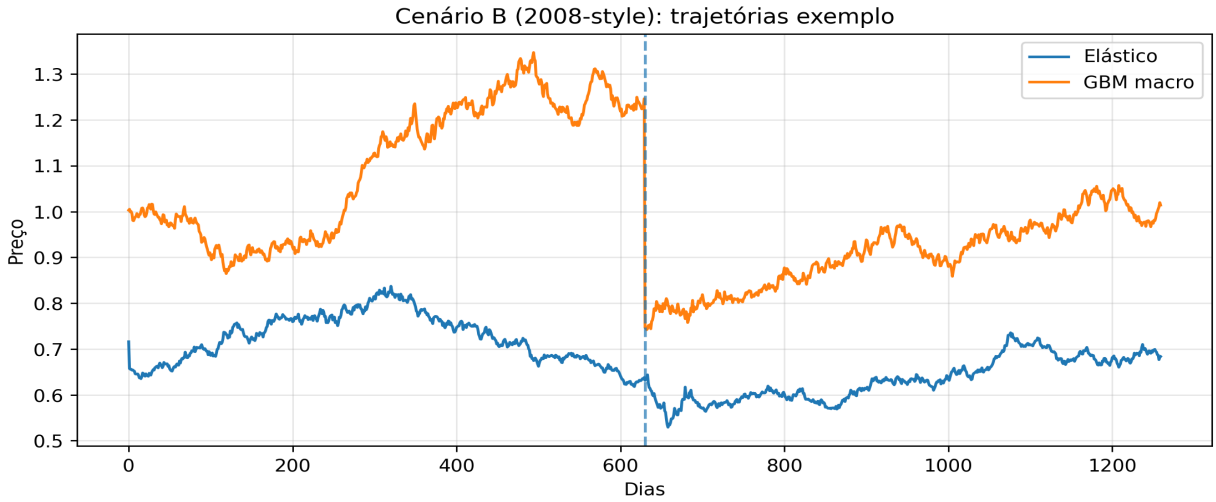

Figure 11. Scenario B (crisis-style): example trajectories.

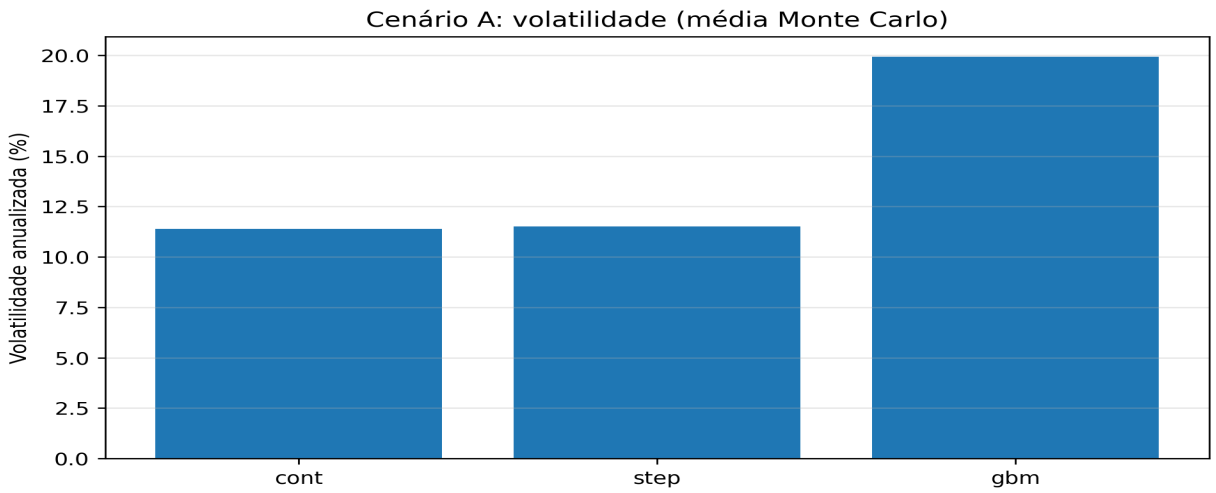

Figure 12. Scenario A: annualized volatility by model (mean Monte Carlo).

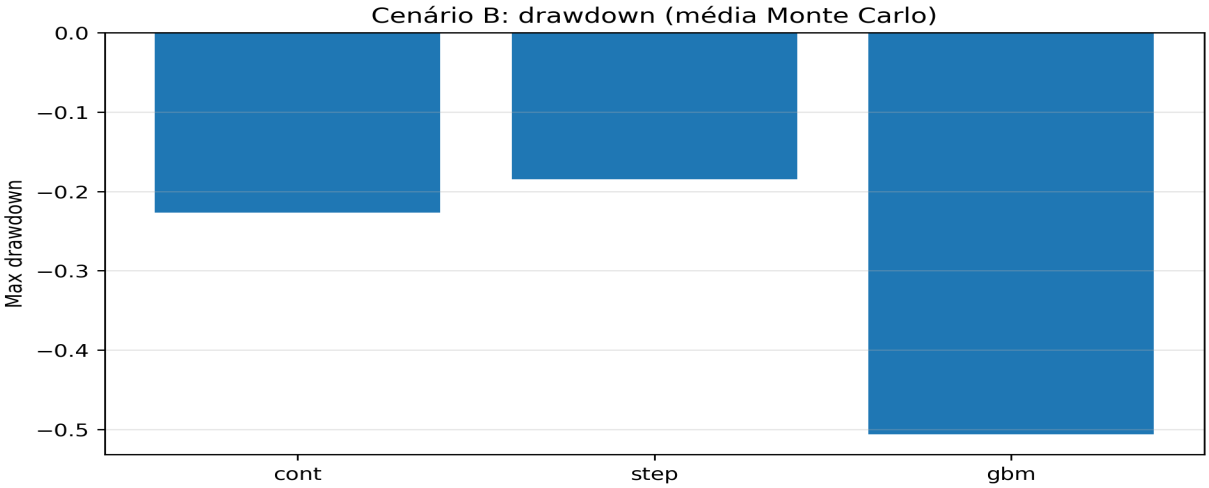

Figure 13. Scenario B: max drawdown by model (mean Monte Carlo).

6. Policy regimes for stepwise supply adjustment (mint/burn)

We extend the stepwise architecture to the supply itself. Under windowed updates, we compare: (i) no burn, (ii) mirror burn proportional to the observed contraction with intensity β in [0,1], and (iii) an amortized rule with a 2% annual structural emission plus partial burn when contraction is detected. This section isolates the trade-off between resilience and policy aggressiveness.

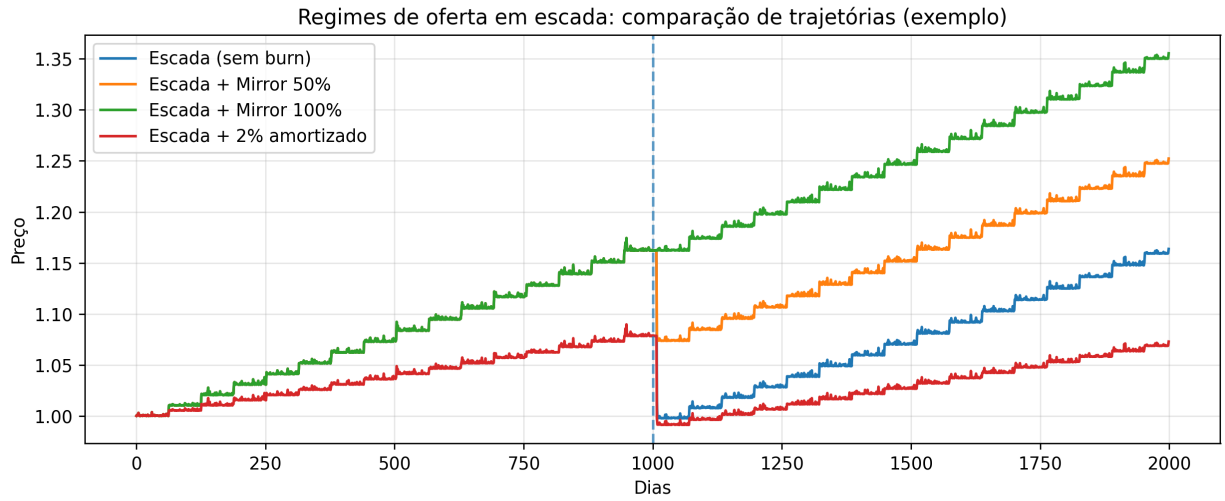

Figure 14. Example trajectories under stepwise supply regimes.

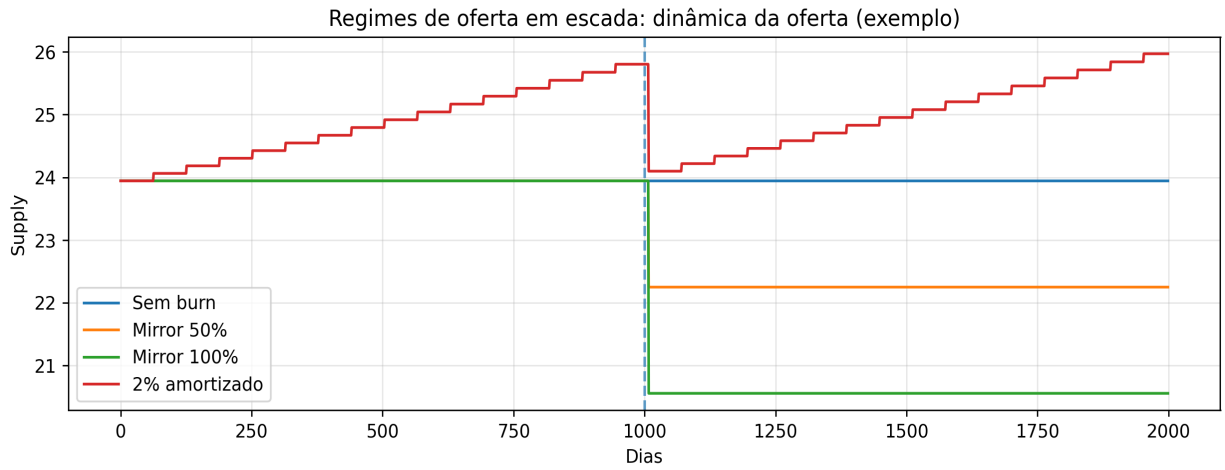

Figure 15. Supply dynamics under stepwise regimes (example).

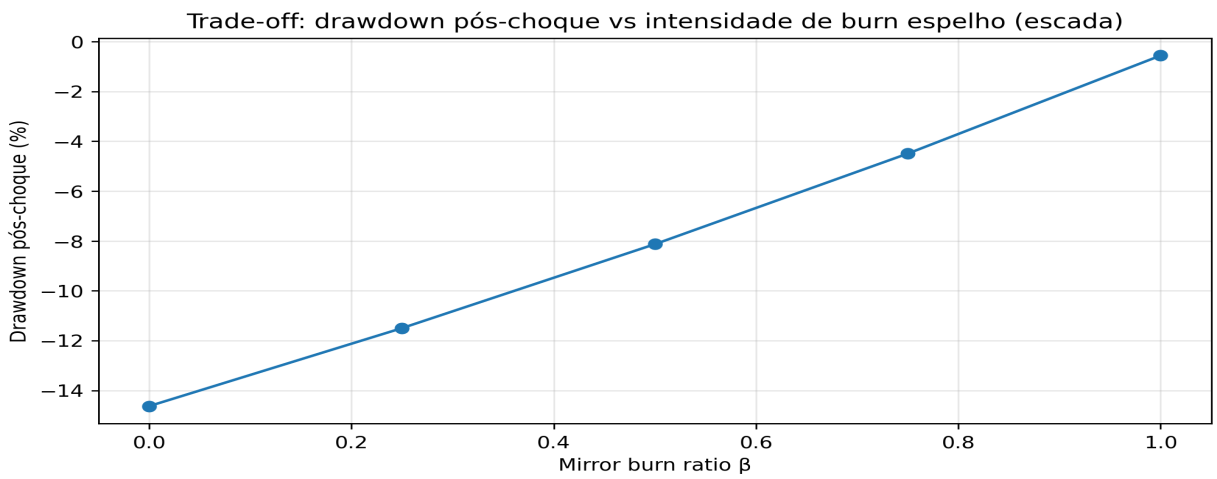

Figure 16. Post-shock drawdown vs. mirror burn intensity β (Monte Carlo).

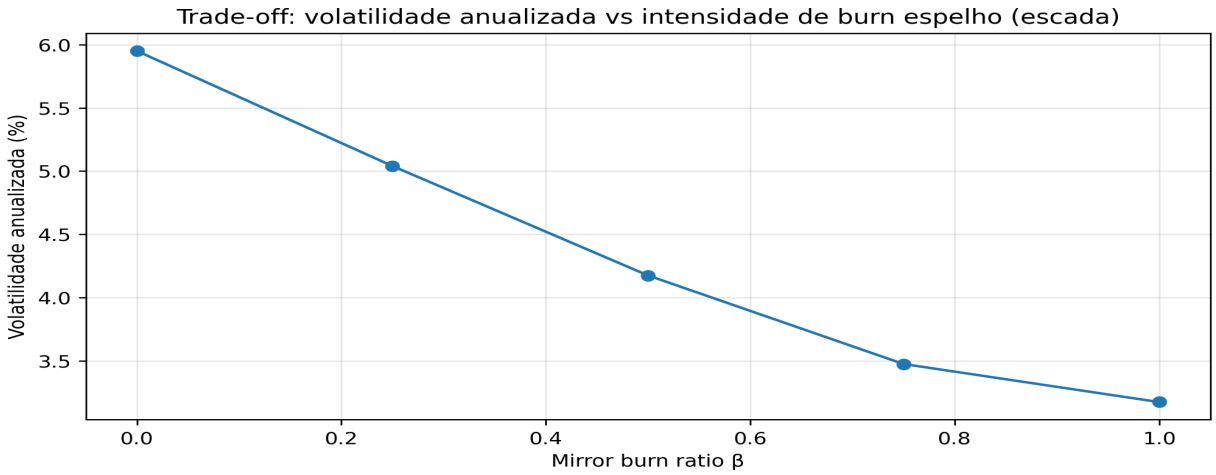

Figure 17. Annualized volatility vs. mirror burn intensity β (Monte Carlo).

Table 2. Summary of stepwise supply regimes (means from Monte Carlo; shock=15%, update=63 days).

Regime	Vol (%)	MDD (%)	DD_shock (%)	Reanchor (d)
Sem burn (β=0)	5.95	-14.64	-14.63	0.0
Mirror β=0.25	5.04	-11.51	-11.50	0.0
Mirror β=0.50	4.18	-8.14	-8.13	0.0
Mirror β=0.75	3.48	-4.51	-4.50	0.0
Mirror β=1.00	3.17	-1.01	-0.56	0.0
2% amortizado (emissão 2% a.a., burn fator=0.5)	3.94	-8.60	-8.59	0.0

7. Clearing-based activity anchors (mini-GDP) with monthly updates

To improve operational feasibility, the model can replace low-frequency macro releases with a high-frequency activity proxy computed from observed intra-bloc clearing volumes. We define a mini-GDP growth signal from clearing volume V_t, smooth it via EMA, and update supply on monthly windows under caps to avoid runaway behavior.

$$g_t^{\text{clear}} = \ln\left(\frac{V_t}{V_{t-1}}\right)$$

$$S_{t+1} = S_t \exp(\beta\, g_t^{\text{EMA}}), \quad |\Delta \ln S| \le \text{CAP}$$

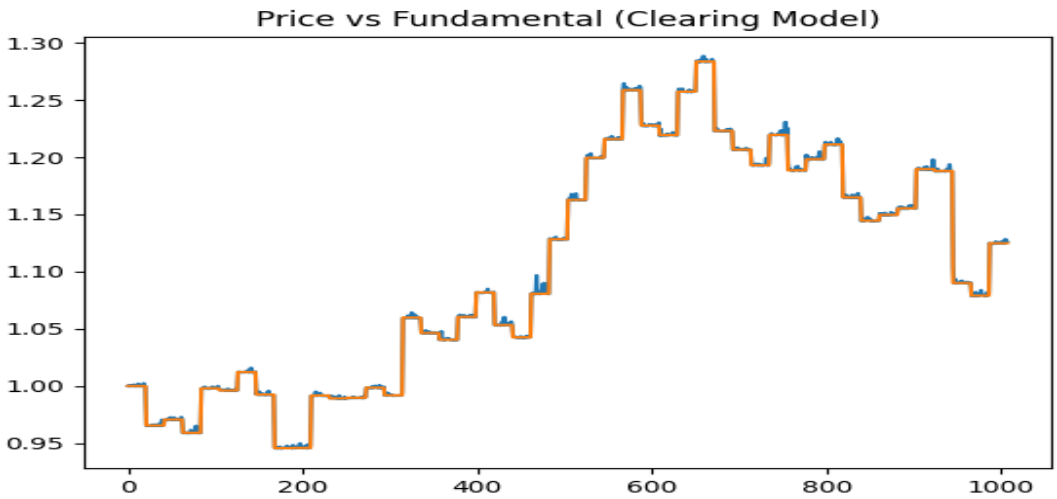

Figure 18. Clearing-based model: price vs. stepwise fundamental under monthly updates (example).

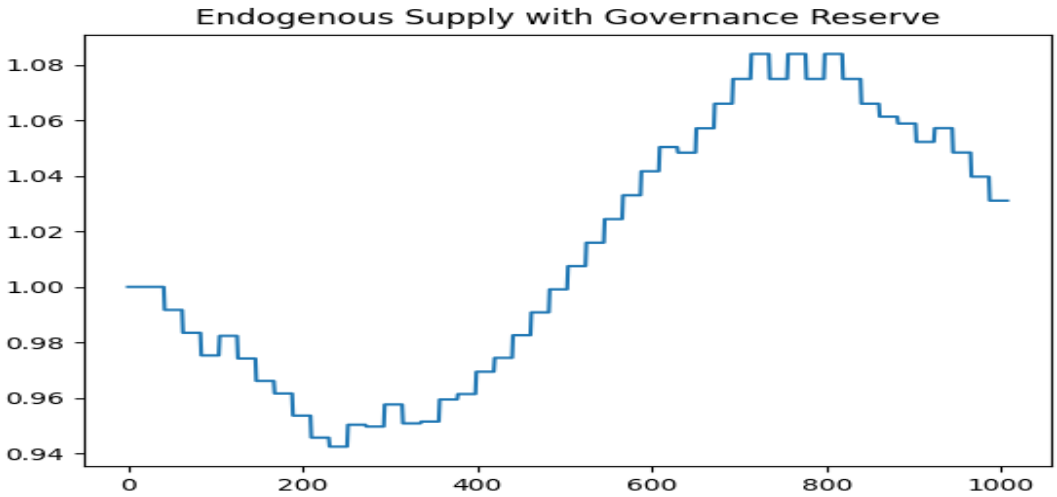

Figure 19. Clearing-based model: endogenous supply path under the mini-GDP rule (example).

8. Institutional implementation: genesis supply and governance reserve

We propose an implementable clearing architecture that avoids political redistribution. A BRICS Clearing Authority (BCA) holds the genesis supply S0 in a reserve account. Members and authorized institutions acquire the unit through demand-based exchange (paying in domestic currency or approved reserve assets), and redeem it symmetrically. Rule-based mint and burn operations, when used, are executed within a governance reserve controlled by the BCA rather than distributed across member states.

Genesis and custody. Total supply S0 is created at inception and held by the BCA reserve account. Member accounts start at zero.

Acquisition/redemption. A member acquires units by transferring domestic currency/reserve assets to the BCA, which transfers units from its reserve. Redemption reverses the operation.

Governance reserve. If the rule increases supply, newly created units accrue to the governance reserve (not to members). If the rule reduces supply, it is burned from that reserve. This avoids implicit cross-country transfers.

9. Long-run drift control

Under structural growth g_real of the reported capital index and structural monetary growth g_monetary (e.g., a fixed emission schedule), the long-run drift of the fundamental satisfies an approximate identity. This provides a simple policy knob for long-run appreciation, neutrality, or inflation regimes.

$$\frac{d}{dt}\ln F_t \approx g_{\text{real}} - g_{\text{monetary}}$$

10. Limitations

The simulations are stylized and intentionally transparent. Capital is proxied as a scaled function of GDP, cross-country correlations are simplified, and market microstructure/liquidity frictions are abstracted. Institutional implementation would require agreed reporting standards for clearing volumes, anti-manipulation controls (e.g., excluding circular transactions), and explicit credit-limit and settlement policies for persistent imbalances.

11. Conclusion

We provide a cohesive design space for a BRICS basket settlement unit: a macro fundamental with elastic reconvergence, a governance dial λ that quantifies concentration, a stepwise reporting layer aligned with real institutional data releases, stress-tested supply policies (mirror burn β and amortized emission), and a clearing-based mini-GDP anchor enabling monthly updates with smoothing and caps. An implementable institutional framework places genesis supply and rule-based mint/burn within a governance reserve held by a clearing authority, avoiding political redistribution while preserving domestic monetary sovereignty.

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